

SUVRADIP GHOSH

West Bengal, India | suvrawork03@gmail.com | +91 8653056408 | linkedin.com/in/suvradip01 | GitHub | Portfolio

SUMMARY

MCA Candidate experienced in building AI-enabled software by combining full-stack development, backend engineering, machine learning, LLMs, computer vision and system optimization to develop applications that solve real-world problems.

SKILLS

Programming Languages: Python, Java, JavaScript, C
Frontend: HTML, CSS, React.js, Tailwind CSS, Redux, Zustand
Backend: Node.js, Express.js, FastAPI
Databases & Cache: MongoDB Atlas, SQL, SQLite, Redis
Tools & Platforms: Git, Docker, Clerk, Convex, ImageKit
AI/ML: PyTorch, TensorFlow, Scikit-Learn, OpenCV, SHAP

PROJECTS

Insight-ATS: AI-Assisted Hiring and Project-Based Evaluation Platform Live | [GitHub](#)

Tech Stack: React.js, Recharts, Redux, FastAPI, Redis, PyTorch, BERT, RoBERTa, DistilBERT, SHAP

- Built an AI-powered ATS platform, **introducing contextual Project Complexity Analysis** and **semantic Resume-JD matching** beyond keyword-based approaches, using **3 transformer models** and SHAP explainability.
- Fine-tuned and evaluated 3 models: **BERT-NER** entity extractor (used 1000 resumes, **0.75 F1**), **RoBERTa** resume-JD semantic matcher (used 8000 resume-JD pairs, **on 0.78 Macro F1**), and **DistilBERT** project-complexity evaluator (**0.93 Macro F1**) using 3×5 repeated K-Fold validation.
- Optimized multi-resume latency by 51.1%** (6.62s to 3.24s) via FastAPI with lazy-loaded singleton orchestration, in-memory model instances, and **Redis caching** (<5ms response for repeat runs).
- Reduced API overhead by 40%** using shared-JD batch screening; utilized **Redux** to manage async uploads and candidate ranking states.

CodeNest: AI-Powered Cloud IDE and Review Platform Live | [GitHub](#)

Tech Stack: React.js, Zustand, Express.js, Redis, MongoDB Atlas, Docker, Groq Llama-3.3-70B

- Built a **microservice-based** cloud IDE supporting **20+ concurrent users**, featuring isolated containerized execution, AI code reviews, flowchart visualization, and **5-second** execution timeouts.
- Engineered a queue-based execution system with a **25-job buffer** and **2 worker processes** to prevent resource exhaustion and ensure reliable execution under concurrent workloads.
- Integrated **Groq Llama-3.3-70B** for ai features and implemented a **Redis-backed AI cache** with in-memory failover, reducing response latency to **1–3 seconds**; secured API endpoints with **SHA-256 cache keys** and **20 req/min** limits.
- Refactored client-side state management using **Zustand** stores with middleware persistence, eliminating prop drilling.

Skin Disease Detection System Using CNN [GitHub](#)

Tech Stack: Python, TensorFlow, EfficientNetB3, OpenCV, Scikit-Learn, CNN

- Developed a **skin disease recognition and classification model** on the HAM10000 dataset (**10,015** dermatoscopic images), restructuring labels into **3 classes**: benign, melanoma, and non-melanoma.
- Prevented data leakage through patient-id based train-validation-test splitting and used a dual-stream preprocessing pipeline using **CLAHE** enhancement and **black-hat morphology**-based hair removal.
- Fine-tuned **EfficientNetB3** using transfer learning, Focal Loss, Adam optimization, and **Test-Time Augmentation**.
- Achieved **90%** test accuracy with F1-scores of **0.89** (benign), **0.86** (melanoma), and **0.94** (non-melanoma), and evaluated performance using confusion matrices and per-class precision/recall analysis.

EDUCATION

Master of Computer Applications Sikkim Manipal Institute of Technology	2024 – Present CGPA: 8.25
Bachelor of Computer Applications Techno India University	2021 – 2024 CGPA: 8.39
Higher Secondary (Class XII) Maynaguri High School	2021 Percentage: 88.6%

CERTIFICATIONS

Android Developer Virtual Internship (Google & AICTE EduSkills) Grade: Excellent	Apr 2024 – Jun 2024 Credential
IBM Professional Certificate: Full Stack Software Developer	Credential
IBM Professional Certificate: Data Science and AI	Credential